

Prob 2: You throw a ball with a speed of 25 m/s at an angle of 40° above the horizontal directly toward a wall which is 22 m apart from the release point of the ball.

- How long does the ball take to reach the wall?
- How far above the release point does the ball hit the wall?
- What are the horizontal and vertical components of its velocity as it hits the wall?
- When it hits the wall, has it passed the highest point of its trajectory?

Solⁿ:

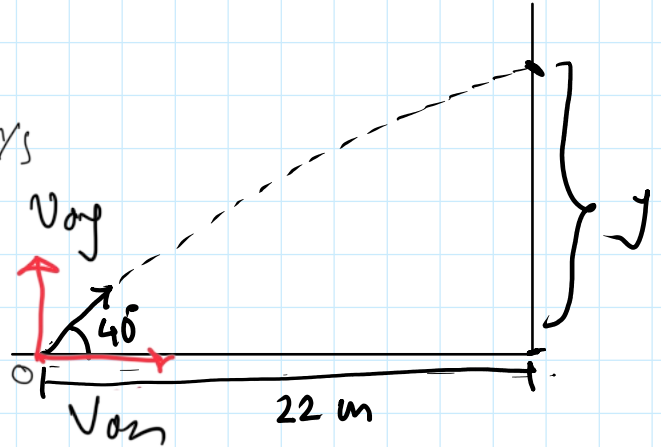
(a) $v_{0x} = 25 \cos 40^\circ = 19.15 \text{ m/s}$

$v_{0y} = 25 \sin 40^\circ = 16.06 \text{ m/s}$

Now, $x = v_{0x} t$

$\Rightarrow t = \frac{x}{v_{0x}} = \frac{22}{19.15} = 1.15$

$\Rightarrow \boxed{t = 1.15 \text{ s}} \text{ Ans}$



(b) $y = y_0 + v_{0y} t + \frac{1}{2} a_y t^2$

$\Rightarrow y = 0 + 16.06 \times 1.15 - 4.9 \times (1.15)^2$

$\Rightarrow \boxed{y = 11.98 \text{ m}} \text{ Ans}$

(c) We have to calculate the velocity at $t = 1.15 \text{ s}$

So, $\boxed{v_x = v_{0x} = 19.15 \text{ m/s}} \text{ Ans}$

$\boxed{v_y = v_{0y} + a_y t = 16.06 - 9.8 \times 1.15 = 4.79 \text{ m/s}} \text{ Ans}$

(d) Since the $v_y = 4.79 \text{ m/s}$, that's why the ball doesn't cross its maximum height.